

SYED SALIK ALVI

Machine Learning & Generative AI Engineer

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SUMMARY

Generative AI engineer specializing in agentic systems. LangChain, LangGraph, LangSmith, and RAG pipelines. Experienced building production grade LLM applications with semantic retrieval, tool calling, and human in the loop workflows. Strong at owning features end to end.

EDUCATION

Sir Syed University of Engineering & Technology

Oct 2022 – Jul 2026

Bachelor of Science in Computer Science | GPA: 3.25 / 4.00 Karachi, Pakistan

Relevant Coursework: Data Structures & Algorithms, Object Oriented Programming, Operating Systems, Database Systems, Software Engineering, Computer Networks, Software Quality Assurance, Artificial Intelligence.

EXPERIENCE

AI / Machine Learning Intern

Aug 2025 – Jan 2026

Dreams Network and Technology Karachi, Pakistan

- Designed and deployed AI driven automation pipelines that streamlined operational workflows across multiple teams, reducing manual processing time by eliminating repetitive data entry tasks.
- Integrated machine learning models into a sales forecasting pipeline using Python, Scikit learn, and SQL backed data processing, improving forecast accuracy through feature engineering and model selection.
- Architected reusable OOP based pipeline modules, cutting integration time for new data sources from days to hours.

Web Development Intern

Aug 2023 – Dec 2023

The Tech Turbine Karachi, Pakistan

- Built responsive frontend features in JavaScript and React using component based design, shipping 8+ UI components adopted across two client projects.
- Refactored REST API consumption layer alongside senior developers, improving average response handling time and reducing duplicated network calls.
- Participated in Agile sprints, code reviews, and Git based collaboration on a codebase shared by a team of 5 developers.

PROJECTS

Text to SQL Agent | Next.js, FastAPI, LangGraph, SQLAlchemy, Groq

[Codebase](#)

- Built a full stack natural language to SQL agent on a LangGraph state machine (inspect schema → generate → validate → execute → format) powered by Qwen3-32B via Groq, querying SQLite, PostgreSQL, and MySQL through a single SQLAlchemy interface behind a Next.js/TypeScript frontend.
- Hardened the agent with three independent read only safety layers (prompt constraints, a SELECT only query validator, and a read only DB connection) and LLM self correction that feeds execution errors back to the model for up to three retries, streaming generated SQL and results live over Server Sent Events.

Real Time NBA Analytics Dashboard | Python, XGBoost, LightGBM, Streamlit, Plotly

[Codebase](#)

- Built an end to end ML system ingesting live NBA data and engineering 12 predictive features across team form, matchup history, and contextual game factors.
- Trained an XGBoost + LightGBM ensemble achieving on game outcome prediction and deployed it via a Streamlit dashboard with auto refresh and interactive Plotly visualizations.

Research Agent | Next.js, FastAPI, LangGraph, LangSmith, Groq, Pinecone

[Codebase](#) · [Live App](#)

- Designed and shipped a full stack agentic RAG research assistant deployed live, with a Next.js/Vercel frontend and a containerized FastAPI backend on Hugging Face Spaces. Built it on an **11 node LangGraph pipeline** (retrieve → evaluate → search → human review → generate) that grounds answers in uploaded PDFs through embedding based semantic retrieval over Pinecone and live web data via Tavily tool calling, with conditional routing and a human in the loop interrupt.
- Delivered by streaming LLM tokens over Server Sent Events for **near instant first token output**. Engineered it to run on free tier infrastructure within **Groq's 6,000 TPM limit with zero hard failures** through context trimming, output caps, and automatic rate limit backoff.

TECHNICAL SKILLS

Languages: Python, C#, JavaScript, SQL, HTML, CSS

Frameworks & Web: .NET, React.js, Next.js, Node.js, FastAPI, REST APIs, Tailwind CSS

Databases & Cloud: Pinecone, ChromaDB, MySQL, SQL Server, Docker, Hugging Face Spaces, Vercel, AWS

GenAI & Agents: LangChain, LangGraph, LangSmith, LlamaIndex, RAG pipelines, Multi agent orchestration, Prompt engineering, Tool / Function calling, Embeddings & semantic search, Model Context Protocol (MCP)

AI / ML & Data: Scikit learn, XGBoost, LightGBM, YOLO (v8 to v11), OpenCV, PyTorch, NumPy, Pandas, Matplotlib, Plotly, Power BI

Concepts: OOP, Data Structures & Algorithms, System Design, RAG, Agent evaluation & observability, Relational Databases, Agile / Scrum, Test Driven Development, Software Quality Assurance

LEADERSHIP & EXTRACURRICULAR

Director of Guest Relations | SSUET Media & Events Council

Nov 2025 – Jan 2026

- Led a team of 12 volunteers coordinating VIP hospitality and protocol logistics for flagship events with 500+ guests.

Corporate Lead | SSUET University Events

Nov 2024 – Jan 2025

- Secured corporate sponsorships for flagship university events, owning outreach, pitching, and negotiation with 20+ companies.